

KRISHNA Koushik Thokala

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Summary

Cloud Infrastructure Engineer building and validating AWS platforms with Terraform, EKS, GitOps, and production-style observability. Implemented Argo CD delivery, GitHub OIDC authentication, Prometheus burn-rate alerting against a 99.9% SLO, autoscaling tests, and failure-recovery drills. AWS Certified Solutions Architect – Associate with an MS in Computer Science and two years of production operations experience leading five-engineer shifts in a 24/7 carrier NOC.

Technical Skills

Cloud & Infrastructure as Code: AWS, Terraform, VPC, EC2, EKS, Lambda, RDS, DynamoDB, S3, API Gateway
Containers & Delivery: Kubernetes, Helm, Docker, ECR, Argo CD, GitOps, GitHub Actions, CI/CD, kubeconform
Security & Access: IAM, OIDC, IRSA, RBAC, KMS, Secrets Manager, Network Policies, Checkov
Observability & Reliability: Prometheus, Grafana, CloudWatch, k6, HPA, SLIs/SLOs, error budgets, burn-rate alerts
Systems & Networking: Python, Bash, SQL, PostgreSQL, Linux, Git, TCP/IP, HTTP/HTTPS, DNS, BGP, MPLS
Certification: AWS Certified Solutions Architect – Associate (July 2026)

Cloud Infrastructure Projects

EKS Reliability Platform | EKS, Terraform, Helm, Prometheus, k6, Python/FastAPI, RDS GitHub

- Provisioned a 59-resource AWS foundation with Terraform: three-tier VPC across public, private, and database subnets, managed EKS with ALB ingress, RDS PostgreSQL, ECR, Secrets Manager, and IRSA-scoped pod IAM roles.
- Implemented six Prometheus availability and latency recording rules with multi-window burn-rate alerts against a 99.9% SLO, validated through five promtool tests.
- Load-tested the FastAPI backend with k6: HPA scaled 2 → 6 replicas across 18,819 requests with zero failed requests or pod restarts, then returned to baseline.

GitOps Delivery Platform | EKS, Argo CD, Terraform, Helm, GitHub Actions, OIDC GitHub

- Composed the AWS foundation as 15 reusable Terraform modules across VPC, EKS, ECR, and IAM under a \$25 monthly AWS Budget guardrail.
- Reconciled four Argo CD Applications across dev, staging, prod, and observability namespaces with per-namespace RBAC, ResourceQuotas, and NetworkPolicies; validated drift self-healing and Git-revert rollback on live EKS.
- Gated image promotion through a four-stage GitHub Actions workflow that verified commit SHA and ECR provenance, required pull-request approval, and authenticated to AWS through branch-scoped OIDC without stored credentials.

Serverless Uptime Monitor | Lambda, EventBridge, DynamoDB, CloudFront/OAC, KMS, X-Ray Live | GitHub

- Deployed a 53-resource serverless stack — scheduled Lambda checks with state-change alerting, API throttling, dead-letter queues, point-in-time recovery, and KMS encryption — running live.
- Modeled a latest-status access pattern that reads 10 current rows instead of scanning 86,400 retained records, holding dashboard read cost flat as retention grows.
- Validated stateful alerting through an outage drill with one-second failure detection, dashboard visibility within 26 seconds, duplicate-alert suppression, and recovery notification.

Experience

Tata Communications July 2022 – July 2024

Senior Engineer, Network Operations Center – Shift Lead

Pune, India

Global telecommunications provider supporting Tier-1 carriers across IP/MPLS and optical transport networks

- Led five-engineer shifts in a 24/7 NOC, prioritizing 40+ daily incidents across 25–30 Tier-1 carrier clients under 99.9% availability and four-hour restoration commitments; trained engineers joining the rotation.
- Owned P1/P2 incidents in ServiceNow, isolating faults across BGP/MPLS, TCP/IP, DNS, DWDM, and hardware while coordinating vendor TAC, field teams, customer testing, and client communication through restoration.
- Executed 4+ planned network changes per month during scheduled maintenance windows, validating service state before and after each change and initiating rollback when verification failed.
- Authored runbooks, post-incident RCAs, and standardized shift handoffs so incoming engineers inherited complete diagnostic context; earned a Certificate of Excellence for resolving an escalated customer case through closure.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering

Chennai, India